

Solution - Assignment-2

Q1 (a)

In Gen. $A = yz\hat{x} + 4xy\hat{y} + y\hat{z}$ at $(1, -2, 3)$

$$\nabla \cdot F = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial (h_2 h_3 F_1)}{\partial u_1} + \frac{\partial (h_1 h_3 F_2)}{\partial u_2} + \frac{\partial (h_1 h_2 F_3)}{\partial u_3} \right]$$

$$(\nabla \cdot A)_R = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$= 0 + 4x + 0 = 4x$$

at $(1, -2, 3)$ $\nabla \cdot A = 4 \leftarrow \underline{\text{ANS}}$

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(b)

$$\nabla \cdot B = \frac{1}{r} \left[\frac{\partial B_r}{\partial r} + \frac{\partial B_\phi}{\partial \phi} + \frac{\partial B_z}{\partial z} \right]$$

$$= \frac{1}{r} \left[\frac{\partial (r \cdot r \sin \phi)}{\partial r} + \frac{\partial (3r^2 \cos \phi)}{\partial \phi} + 0 \right]$$

$$= \frac{1}{r} [2r \sin \phi - 3r^2 \sin \phi]$$

$$= (2 - 3r) \sin \phi$$

put $r = 1$ & $\phi = 90^\circ$

$\nabla \cdot B$ at given point $= (2 - 3 \times 1) \cdot 1 \cdot \sin 90^\circ = -1$ ANS
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(c) $\nabla \cdot C = \frac{1}{r^2 \sin \theta} \left[\frac{\partial (r \cdot r \sin \theta \cdot 2r \cos \theta \cos \phi)}{\partial r} + \frac{\partial (r \sin \theta \cdot r^2)}{\partial \theta} \right]$

$$= \frac{1}{r^2 \sin \theta} [2 \times 3 r^2 \sin \theta \cos \theta \cos \phi + 0]$$

$$= 6 \cos \theta \cos \phi, \leftarrow \text{put } \theta = \pi/6, \phi = \pi/3$$

$(\nabla \cdot C) = 6 \times \cos(\pi/6) \cdot \cos(\pi/3) = 2.598 \underline{\text{Ans}}$

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Assignment-2

Sol: Q2

$$(a) f = x^2 + y^3 + z^4$$

$$\begin{aligned}\nabla f &= \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z} \\ &= 2x \hat{x} + 3y^2 \hat{y} + 4z^3 \hat{z} \leftarrow \text{ANS.}\end{aligned}$$

(b) Gen. Exp. of ∇f

$$\nabla f = \frac{1}{h_1} \frac{\partial f}{\partial u_1} \hat{u}_1 + \frac{1}{h_2} \frac{\partial f}{\partial u_2} \hat{u}_2 + \frac{1}{h_3} \frac{\partial f}{\partial u_3} \hat{u}_3$$

$$\nabla f = \frac{1}{r} \frac{\partial (f \cos \theta)}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial (f \cos \theta)}{\partial \theta} \hat{\theta} + \frac{1}{r} \frac{\partial (f \cos \theta)}{\partial \phi} \hat{\phi}$$

$$\nabla f = \cos \theta \hat{r} - f \sin \theta \hat{\theta} \leftarrow \text{ANS.}$$

(c)

$$\nabla f = \frac{\partial (r \sin \theta \sin \phi)}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial (r \sin \theta \sin \phi)}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial (r \sin \theta \sin \phi)}{\partial \phi} \hat{\phi}$$

$$\nabla f = \sin \theta \sin \phi \hat{r} + \cos \theta \sin \phi \hat{\theta} + \cos \phi \hat{\phi} \leftarrow \text{ANS.}$$

Assignment_2

Sol. Q3

(a) V_{MN} if point M & N are $M(2, 6, -1)$, $N(-3, -3, 2)$

$$\begin{aligned} V_{MN} &= -\int E \cdot dl = -\left[6 \int_{-3}^2 x^2 dx + 6 \int_{-3}^6 y dy + 4 \int_{-1}^2 dz \right] \\ &= -\left[\frac{6}{3} (x^3)_{-3}^2 + \frac{6}{2} (y^2)_{-3}^6 + 4 (z)_{-1}^2 \right] \\ &= -\left[2 \{ (2)^3 - (-3)^3 \} + 3 \{ (6)^2 - (-3)^2 \} + 4 \{ -1 - 2 \} \right] \\ &= -[70 + 81 - 12] = -139 V \leftarrow \underline{\underline{ANS}} \end{aligned}$$

(b) V_m if $\overline{V=0}$ at $Q(4, -2, -35)$

$$\begin{aligned} V_{mQ} &= -\left[6 \int_4^2 x^2 dx + 6 \int_{-2}^6 dy + 4 \int_{-35}^{-1} dz \right] \\ &= -\left[2 \{ x^3 \}_4^2 + 3 \{ y^2 \}_{-2}^6 + 4 \{ z \}_{-35}^{-1} \right] \\ &= -\left[2 \{ (2)^3 - (4)^3 \} + 3 \{ (6)^2 - (-2)^2 \} + 4 \{ -1 + 35 \} \right] \\ V_{mQ} = V_m - V_Q &= -[-112 + 96 + 136] = -120 V \leftarrow \underline{\underline{ANS.}} \end{aligned}$$

(c) $\overline{V_m = -120V}$

$N(-3, -3, 2)$, $P(1, 2, -4)$, $V_P = 2V$ (given)

$$V_{NP} = -\left[6 \int_1^{-3} x^2 dx + 6 \int_2^{-3} y dy + \int_2^{-4} 4 dz \right]$$

$$= -\left[\frac{6}{3} (x^3)_1^{-3} + \frac{6}{2} (y^2)_2^{-3} + 4 (z)_{-4}^2 \right]$$

$$= -[-56 + 15 + 24] = 17$$

$$V_{N-2} = 17 \therefore V_N = 17 + 2 = \underline{\underline{19V}} \leftarrow \underline{\underline{ANS}}$$

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Assignment-2

Sol: Q4

$P(0.2, -0.4, 0.4)$

(a) $V_{PQ} = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_P} - \frac{1}{r_Q} \right)$, r_P & r_Q are distances of points P & Q resp. from the charge (source) which is at origin $(0, 0, 0)$

$$r_P = \sqrt{(0.2)^2 + (-0.4)^2 + (0.4)^2} = 0.6$$

$$r_Q = \infty$$

$$V_{PQ} = \frac{6 \times 10^{-9}}{4\pi\epsilon_0} \left(\frac{1}{0.6} - \frac{1}{\infty} \right) = 89.877 \text{ V}$$

Now $V_{PQ} = V_P - V_Q$ & $V_Q = 0$ (given)

$$\therefore V_P = V_{PQ} + V_Q = 89.877 + 0 = \underline{\underline{89.877 \text{ V}}} \quad \underline{\underline{\text{ANS}}}$$

(b) Let Q is $(1, 0, 0)$

then $r_Q = \sqrt{(1)^2 + 0^2 + 0^2} = 1$

$$V_{PQ} = \frac{6 \times 10^{-9}}{4\pi\epsilon_0} \left(\frac{1}{0.6} - \frac{1}{1} \right) = 35.951 \text{ V}$$

$$V_{PQ} = V_P - V_Q \text{ and } V_Q = 0$$

$$\therefore V_P = 35.951 \text{ V} \leftarrow \underline{\underline{\text{ANS}}}$$

(c) Let Q is $(-0.5, 1, 1)$, then

$$r_Q = \sqrt{(-0.5)^2 + (1)^2 + (1)^2} = 1.5$$

$$V_{PQ} = V_P - V_Q \text{ and } V_Q = 20 \text{ V (given)}$$

$$V_P = V_{PQ} + V_Q = \frac{6 \times 10^{-9}}{4\pi\epsilon_0} \left(\frac{1}{0.6} - \frac{1}{1.5} \right) = 53.926 \text{ V}$$

$$V_P = V_{PQ} + V_Q = 53.92 + 20 = \underline{\underline{73.926 \text{ V}}} \leftarrow \underline{\underline{\text{ANS}}}$$

Assignment-2

Sol: Q5:

(a) $\vec{p} = Q\vec{d}$ $\vec{d} \rightarrow$ Vector from -ve charge towards +ve charge

$$\vec{d} = [0.001\hat{z} - (-0.001\hat{z})] = 0.002\hat{z} \text{ m}$$

$$\vec{p} = 3 \times 10^{-6} (0.002\hat{z}) = 6 \times 10^{-9} \hat{z} \text{ C-m}$$

$$= \underline{6 \hat{z} \text{ nC-m}} \quad \text{ANS}$$

(b) Since dipole centre is located at origin we may use

$$\vec{E} = \frac{Qd}{4\pi\epsilon_0 r^3} (2\cos\theta\hat{r} + \sin\theta\hat{\theta})$$

$$= \frac{3 \times 10^{-6} \times 0.002}{4\pi\epsilon_0 (2)^3} \times (2\cos(40^\circ)\hat{r} + \sin 40^\circ\hat{\theta})$$

$$= 10.328\hat{r} + 4.333\hat{\theta} \text{ V/m}$$

(c) At point (1, 2, 1.5) we have

$$r = \sqrt{(1)^2 + (2)^2 + (1.5)^2} = 2.6926$$

$$\theta = \cos^{-1}(1.5/2.69258) = 56.145^\circ$$

$$\phi = \tan^{-1}(2/1) = 63.435^\circ$$

$$\vec{E} = \frac{3 \times 10^{-6} \times 0.002}{4\pi\epsilon_0 (2.69258)^3} (2\cos 56.145^\circ\hat{r} + \sin 56.145^\circ\hat{\theta})$$

$$\vec{E} = 3.078\hat{r} + 2.294\hat{\theta} \text{ V/m}$$

$\leftarrow$ ANS.